HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME		**CT GROUP**	**20S7__**
CENTRE NUMBER		**INDEX NUMBER**	

BIOLOGY **9744/02**

Paper 2 Structured Questions **31 August 2021**

Candidates answer on the Question Paper. **2 hours**

No Additional Materials are required.

INSTRUCTIONS TO CANDIDATES

Write your **name**, **CT group, Centre number** and **index number** in the spaces provided at the top of this cover page.

There are **nine** questions.

Answer **all** questions in the spaces provided on the Question Paper.

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	**/ 10**
2	**/ 10**
3	**/ 10**
4	**/ 11**
5	**/ 11**
6	**/ 9**
7	**/ 13**
8	**/ 13**
9	**/ 13**
Total	**/ 100**

This document consists of **25** printed pages.

QUESTION 1

Sucrose phosphorylase is found in some bacteria. It exists as a dimer around the active site, which has two binding sites, one for the sugar molecule and the other for water molecule.

Fig. 1.1 shows structure of sucrose phosphorylase.

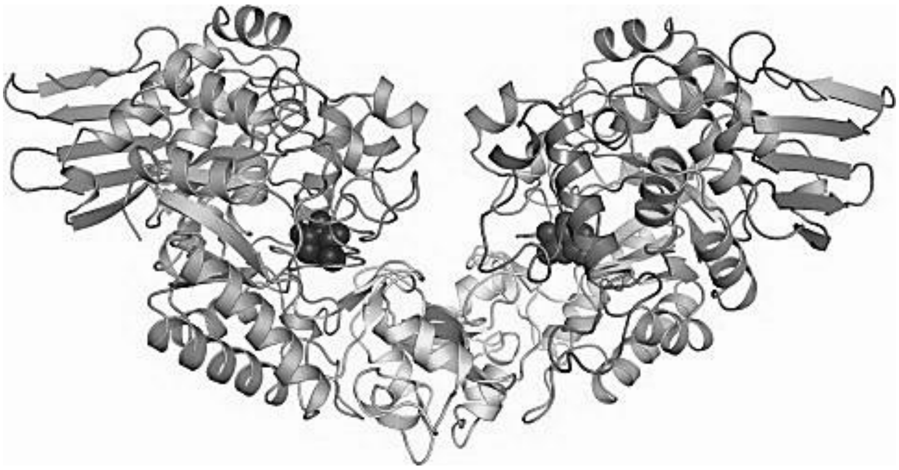

Fig. 1.1

(a)(i) State the highest level of protein structure observed in Fig. 1.1.

.... [1]

(ii) Describe how this structure is maintained.

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.... [3]

(iii) A molecule of water binds to the active site of sucrose phosphorylase.

Explain the role of the water molecule in the active site.

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.... [2]

An enzyme that catalyses a reaction of commercial interest needs to be investigated to see if it is suitable for use in industry.

For example:

- immobilised enzymes may be used as they have a longer shelf-life than the free enzyme in solution
- many industrial reactions are carried out at higher temperatures to minimise contamination of products by microorganisms.

Fig. 1.2 shows the results of an investigation to compare the activity of sucrose phosphorylase free in solutions (free enzyme) with immobilised sucrose phosphorylase (immobilised enzyme) at different pHs.

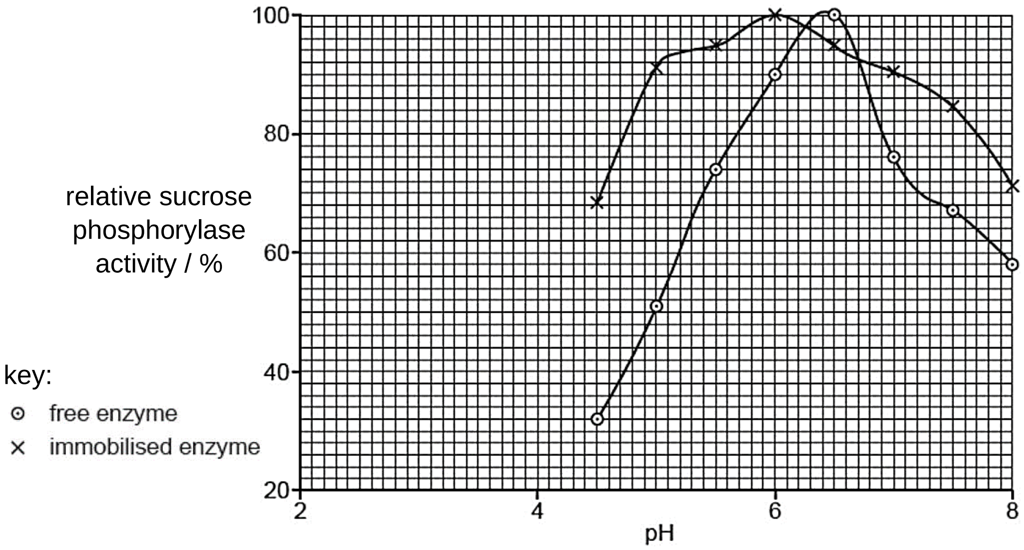

Fig. 1.2

Fig. 1.3 shows the activity of the free enzyme and immobilised enzyme at different temperatures.

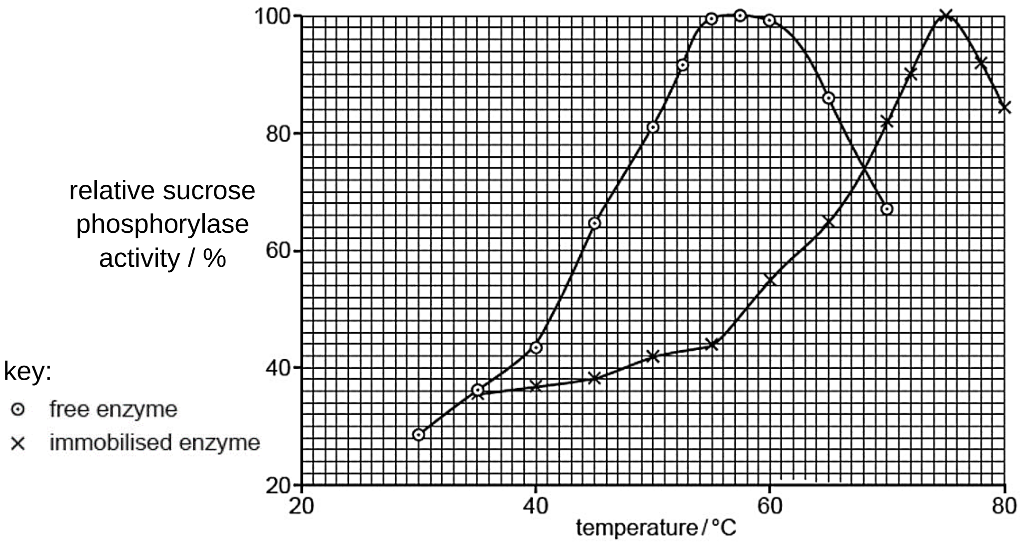

Fig. 1.3

(b) With reference to the results shown in Fig. 1.2 and Fig. 1.3, discuss which sucrose phosphorylase enzyme, free or immobilised, is better for use in industrial reactions.

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.... [4]

[Total: 10]

QUESTION 2

The cell membrane has a fluid mosaic structure.

(a) Describe what is meant by the term *fluid mosaic*.

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.... [2]

In 1935, biologists Davson and Danielli published their suggestion for the structure of cell membranes.

This was later updated with the model proposed by Singer and Nicolson in 1972.

Both models are shown in Fig. 2.1.

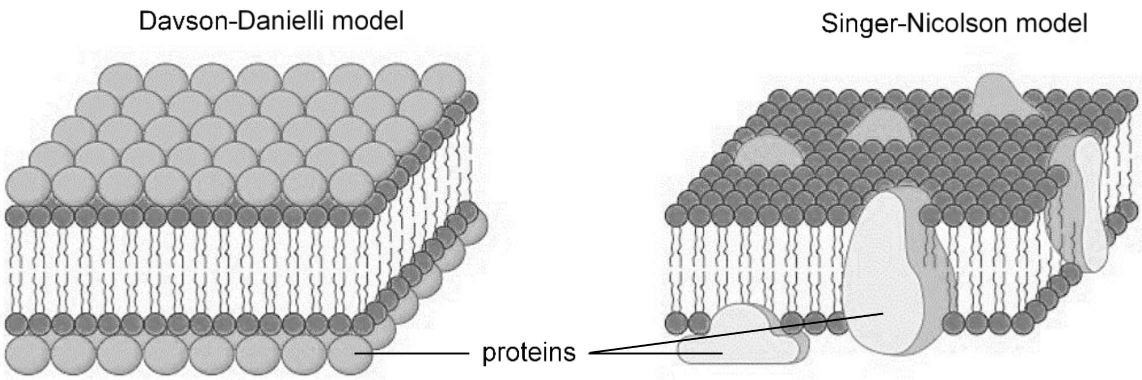

Fig. 2.1

(b) Distinguish between these two models.

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.... [2]

Membrane integrity is essential for the proper movement of molecules across a cell membrane.

(c) Explain how high temperatures could damage the structure of cell membranes.

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.... [4]

Testosterone is a steroid hormone that enters the cell naturally.

(d) Suggest how the transport of testosterone into the cell would be affected by high temperatures.

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.... [2]

[Total: 10]

QUESTION 3

(a) Account how **one** structural feature of DNA contribute to its stability as a hereditary material.

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.... [2]

The “trombone” model of DNA replication postulates that two DNA polymerase enzymes work together in a protein complex during DNA replication.

Fig. 3.1 shows a DNA molecule undergoing DNA replication.

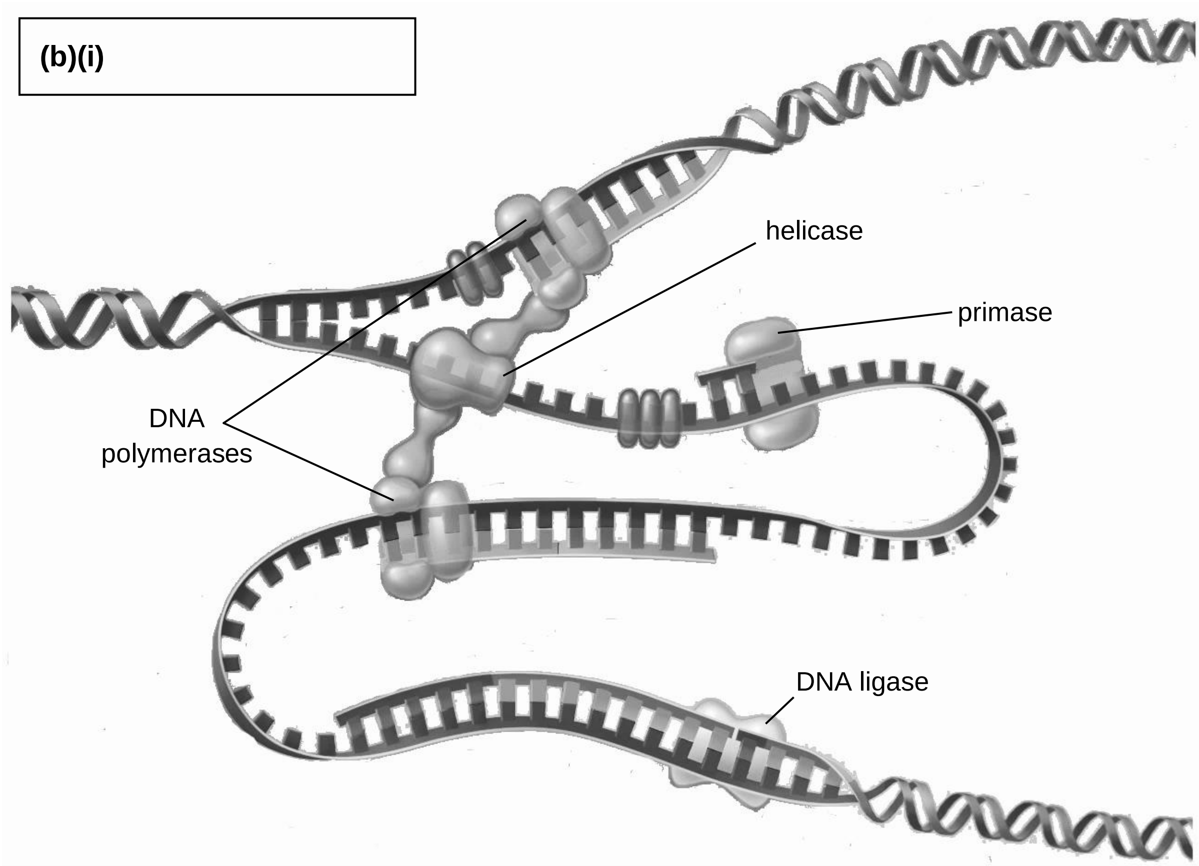

Fig. 3.1

(b) With reference to Fig. 3.1,

(i) indicate the overall direction of the replication fork in the box provided. [1]

(ii) compare how the synthesis of the lagging strand differs from that of the leading strand.

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.... [2]

(iii) suggest why it is necessary for two DNA polymerase enzymes to work together in a protein complex during DNA replication.

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.... [2]

(c) Explain how the end replication problem arises.

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.... [3]

[Total: 10]

QUESTION 4

(a) Explain how the cell theory applies to prokaryotes.

[2]

Prokaryotic and eukaryotic cells differ significantly in many aspects.

(b) Distinguish between the structure of the prokaryotic and eukaryotic genomes.

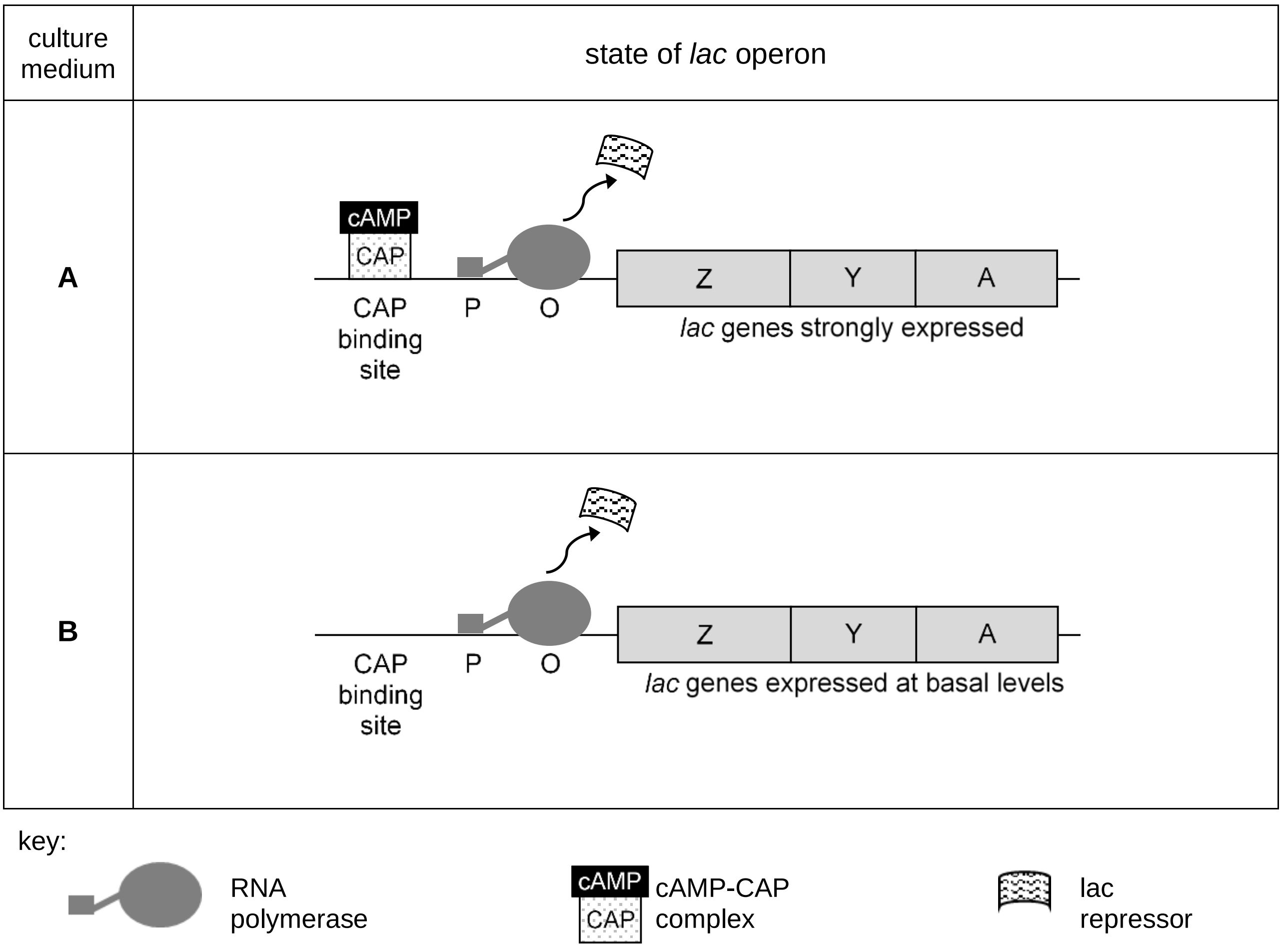
[2]

The prokaryotic genome can be organised into operons. The *lac* operon is one such example.

(c) Describe the structural features of the mRNA transcribed from the *lac* operon.

[2]

Fig. 4.1 shows the state of *lac* operon in bacteria grown in two different culture media, **A** and **B**.



Note: Fig. 4.1 is not drawn to scale

Fig. 4.1

(d) Using a tick (✓) or a cross (×), indicate the presence or absence of each of the following metabolites in culture medium **A**. [1]

metabolite	presence or absence of metabolite
glucose	
lactose	

(e) State **and** explain how the composition of culture medium **B** results in the basal expression of the *lac* operon.

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.... [4]

[Total: 11]

QUESTION 5

To investigate the effects of distal control elements on gene expression, a researcher constructed three different artificial gene sequences shown in Fig. 5.1.

In these sequences, the various control elements were fused with the luciferase gene, also known as a reporter gene. The three different gene sequences were then introduced into mammalian cells separately.

When the luciferase gene is expressed, it produces a bioluminescent signal when the substrate luciferin is added onto the cells. A luminometer can be used to read the bioluminescent signals.

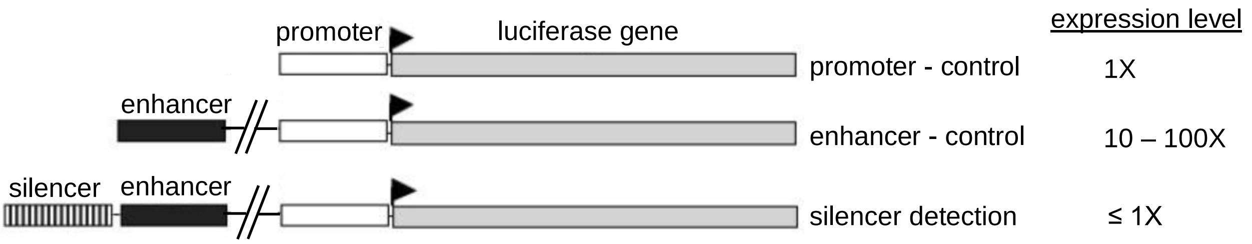

Fig. 5.1

(a) Explain the role of the luciferase gene in this investigation.

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.... [2]

(b) Comment on how these results illustrate the regulation of luciferase gene expression.

[6]

(c) Discuss the significance of the specific combinations of control elements and transcription factors in regulating gene expression.

[3]

[Total: 11]

QUESTION 6

Two groups of genes are involved in regulating cell division: proto-oncogenes and tumour suppressor genes.

The *K-ras* gene encodes the protein KRAS, which is a GTPase that acts as a molecular switch. Genetic alteration of *K-ras* gene has been found to be frequently associated with pancreatic cancer.

(a) Identify and explain to which group of genes *K-ras* belong to in regulating cell division.

[3]

(b) Discuss how a genetic change to the *K-ras* gene leads to excessive cell proliferation.

[4]

Early detection of pancreatic cancer remains a challenge even though a person has developed the condition. A large proportion of patients is diagnosed only when the cancer is in the advanced stage.

(c) Suggest why pancreatic cancer remains a challenge to be diagnosed early.

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.... [2]

[Total: 9]

QUESTION 7

Mendelian inheritance refers to an inheritance pattern that follows Mendel's First Law of Segregation and Mendel's Second Law of Independent Assortment.

However, not all genes are inherited by following the Mendelian laws. Linked genes are an example.

In rabbits, the black allele **B** is dominant to the brown allele **b**, while the full colour allele **H** is dominant to the chinchilla allele **h**.

A researcher crossed a pure-breeding black, chinchilla rabbit with another pure-breeding brown, full colour rabbit. The resultant F_1 offspring are all black, full colour rabbits.

All of these F_1 offspring were then crossed with brown, chinchilla rabbits, in a series of crosses, producing the following 100 progeny:

31 black, chinchilla
34 brown, full colour
16 black, full colour
19 brown, chinchilla

(a)(i) State Mendel's Second Law of Independent Assortment.

....

.... [1]

(ii) State what is meant by *pure-breeding* in this context.

....

.... [1]

(b) Draw a genetic diagram to show the expected phenotypic ratio for the progeny resulting from the series of crosses between the F_1 offspring and brown, chinchilla rabbits if inheritance is Mendelian.

[5]

(c) Explain how it is then possible to obtain the observed numbers in the 100 progeny from the series of crosses between the F_1 offspring and brown, chinchilla rabbits.

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.... [3]

(d) Suggest how researchers can use similar breeding experiments with many different pairs of characters to map the position of genes on the chromosomes of rabbits.

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.... [3]

[Total: 13]

QUESTION 8

Coenzyme A (CoA) is a nucleotide that plays an important role in cellular respiration.

Fig. 8.1 represents the molecular structure of coenzyme A.

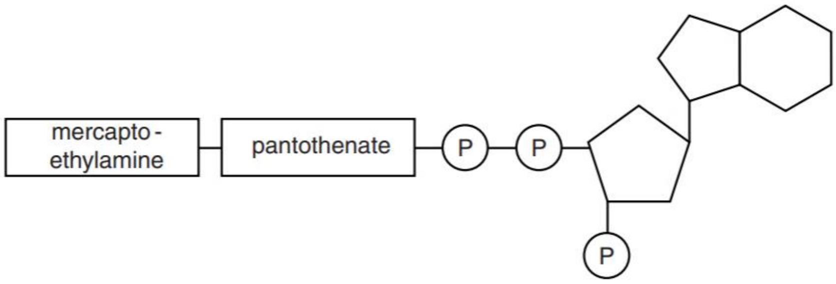

Fig. 8.1

(a) State two structural features of nucleotides that are visible in coenzyme A.

feature 1 ....

feature 2 ....

[2]

Fig. 8.2 is a diagram of the link reaction and Krebs cycle, which are key stages in cellular respiration.

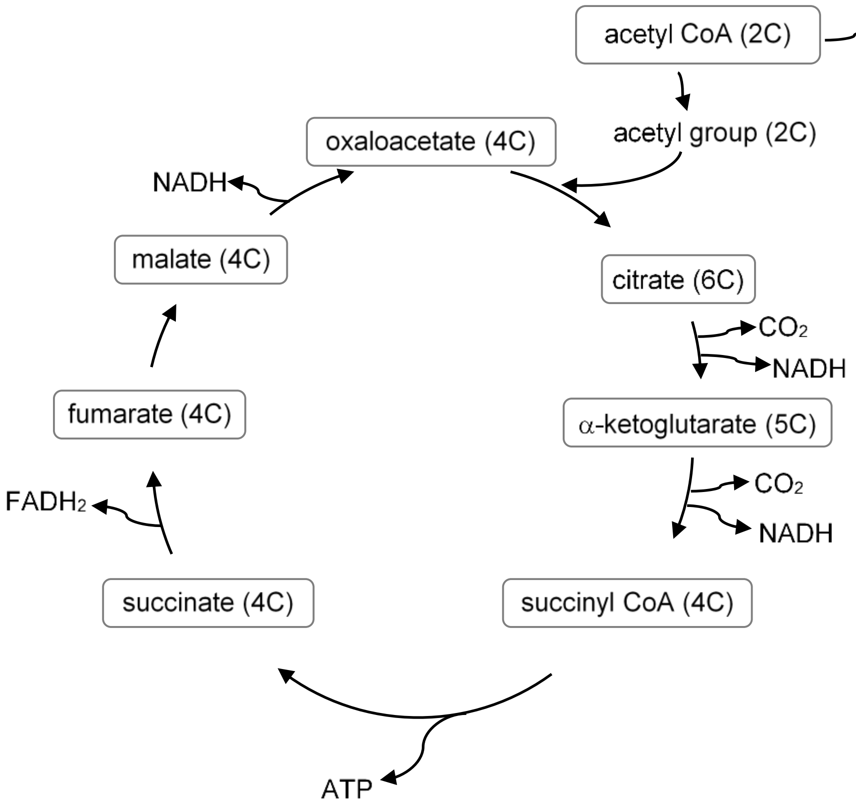

Fig. 8.2

(b) Describe the role of coenzyme A in cellular respiration.

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.... [2]

(c) Outline how ATP is synthesised in Krebs cycle.

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.... [2]

Fluoroacetate, a two-carbon (2C) compound originally isolated from plants, is a potent toxin.

The toxic effect of fluoroacetate was studied in an experiment using rat muscle tissue.

- After the muscle tissue was incubated with fluoroacetate, a six-carbon (6C) compound, fluorocitrate, was detected.
- The concentration of most Krebs cycle intermediates were below normal, except for six-carbon (6C) citrate, with a concentration ten times higher than normal.

(d) With reference to Fig. 8.2,

(i) explain how fluoroacetate inhibits the Krebs cycle.

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.... [3]

(ii) suggest how the inhibition can be overcome.

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.... [1]

(e) Explain why the administration of fluoroacetate resulted in no ATP synthesis.

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.... [3]

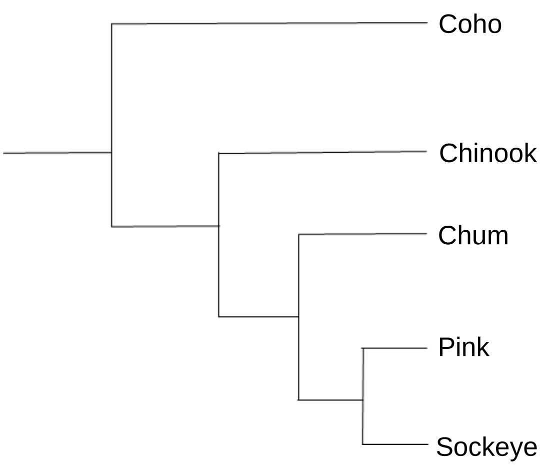
[Total: 13]

QUESTION 9

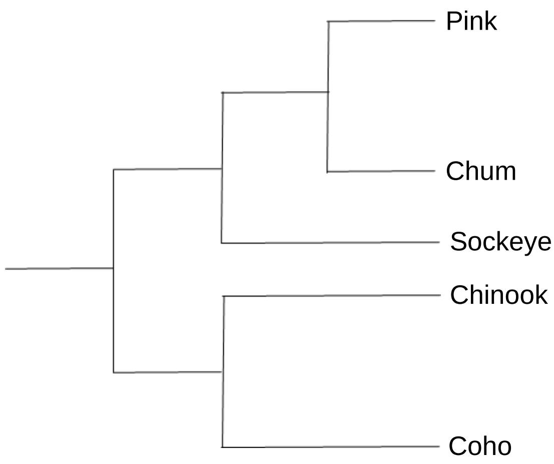
The *Oncorhynchus* genus of fish contains five species of Pacific Salmon residing in the Pacific Ocean.

Fig. 9.1 shows two different ways of classifying the same five species of Pacific Salmon where

- classification **X** is based on the frequency of morphological characteristics
- classification **Y** is based on other comparisons of genetic characteristics.



classification **X**



classification **Y**

Fig. 9.1

(a)(i) Describe **two** differences in the evolutionary relationships amongst these five species of Pacific Salmon based on the two classifications shown in Fig. 9.1.

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.... [2]

(ii) State **one** reason why classification **Y** is a better representation of evolutionary relationships than classification **X**.

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.... [1]

The coho salmon, *Oncorhynchus kisutch*, spend approximately equal time in fresh water streams, where the young fish mature in stages, and in the salt water of the Pacific Ocean, where adults spend their majority of their time.

Adults migrate back to the stream in which they hatched from eggs (natal stream) in order to spawn (release sperm or eggs).

Some males, known as jacks, begin the migration to the natal stream much earlier in their adult lives than the normal breeding adult males, known as hooknoses.

Fig. 9.2 depicts hooknoses and jacks. Jacks are reproductively mature but are much smaller than hooknoses and do not develop the hooked snout and large teeth.

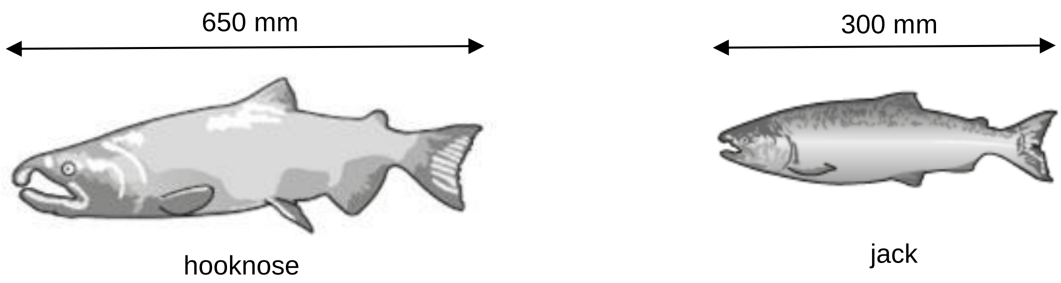

Fig. 9.2

Jacks and hooknoses employ different breeding strategies in order to spawn successfully.

- Jacks sneak around the smaller boulders on the stream bed and attempt to stealthily mate with a female.
- Hooknoses swim within the open water and fight aggressively amongst one another for the opportunity to mate with a female.

(b) Complete Table 9.1 to show the classification of coho salmon.

Table 9.1

taxon	coho salmon
kingdom	Animalia
phylum	Chordata
	Actinopterygii
	Salmoniformes
family	Salmonidae
genus	*Oncorhynchus*

[2]

(c) Suggest reasons why jacks are able to spawn as successfully as hooknoses.

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.... [2]

Fig. 9.3 shows the variation in the body length of reproductively mature males in an original population of coho salmon, before evolution of the jack reproductive phenotype.

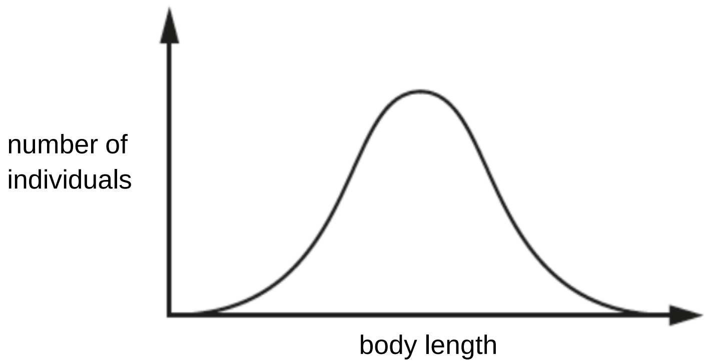

Fig. 9.3

Selection has since acted over time on this original population to change the distribution of body length in reproductively mature male salmon.

(d)(i) Sketch a curve on Fig. 9.3 to show the new distribution of body length in the present-day population of reproductively mature male salmon. [1]

(d)(ii) Explain the type of selection that is occurring.

[2]

Conservation strategies have been put in place for several populations of coho salmon, which are listed as endangered.

One population, the Interior Fraser population, is genetically distinct as all its closely related populations are extinct. Many of its sub-populations, originating from the small rivers flowing into the main river Fraser, have experienced a severe decrease in numbers in recent years.

(e) Explain why the loss of sub-populations of the Interior Fraser population represents a decrease in biodiversity.

[3]

[Total: 13]

--- END OF PAPER---

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